

CSE106 Discrete Mathematics

Credit Hours: $3 + 0 = 3$

Prerequisite: CSE103 Structured Programming

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Text book:

Kenneth H. Rosen, *Discrete Mathematics and Its Applications with Combinatorics and Graph Theory*, 7th Edition, Tata McGraw-Hill Publishing Company Limited, New Delhi, 2015.[5th and 6th Edition will also be sufficient]

Midterm I

Link:

<https://drive.google.com/file/d/1MtGld11D10CHWk1VXwXIIW4ciE9Qb1ue/view?usp=sharing>

Chapter 1

The Foundations: Logic and Proofs

1.1 Propositional Logic

Proposition:

A proposition is a *declarative sentence* that is either **True** or **False**, but not both.

Example 1: Following declarative sentences are propositions

1. Dhaka is the capital of Bangladesh (True)
2. Rajshahi is the capital of Bangladesh (False)
3. $1 + 1 = 2$ (True)
4. $2 + 2 = 3$ (False)

Example 2: Following sentences are not propositions

1. What time is it? (Not a declarative sentence)
2. Read this carefully. (Not a declarative sentence)
3. $x + 1 = 2$ (A declarative sentence but the truth value can be determined when the value of x is known)
4. $x + y = z$ (A declarative sentence but the truth value can be determined when the values of x , y , and z are known)

Propositional variables:

Variables represent propositions. The conventional letters used as propositional variables p, q, r, s , etc.

Truth value of a proposition:

Truth value of a proposition is True (T) if it is a true proposition and False (F) if it is a false proposition.

Compound propositions:

Compound propositions are formed from existing propositions using **logical operators**.

Definition 1:

Negation or NOT operator ($\neg$): Changes the truth value of the proposition.

Example 3:

p : Today is Friday. (False)

$\neg p$: Today is **not** Friday. (True)

Example 4:

p : At least 10 inches of rain fell today in Dhaka. (False)

$\neg p$: Less than 10 inches of rain fell today in Dhaka. (True)

Table 1: The Truth Table for the Negation (NOT) of a Proposition.

p	$\neg p$
T	F
F	T

Connectives:

The logical operators that are used to form new propositions from two or more existing propositions are called connectives.

Definition 2:

Conjunction or AND operator ($\wedge$): The conjunction $p \wedge q$ is True when both p and q are true and False otherwise.

Example 5:

p : Today is Friday.

q : It is raining today.

$p \wedge q$: Today is Friday **and** it is raining today.

It is True on Friday when it is raining (rainy Friday).

It is False on days other than Friday.

It is False on days that are not raining.

It is False on Fridays when it is not raining.

Table 2: The Truth Table for the Conjunction (AND) of Two Propositions.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Definition 3:

Disjunction or OR operator ($\vee$): The disjunction $p \vee q$ is False when both p and q are False and True otherwise.

Example 6:

p : Today is Friday.

q : It is raining today.

$p \vee q$: Today is Friday **or** it is raining today.

It is True on Friday.

It is True on all rainy days.

It is True on Friday when it is raining.

It is False on days other than Friday when it is not raining.

Table 3: The Truth Table for the Disjunction (OR) of Two Propositions.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Definition 4:

Exclusive OR operator ($\oplus$): The *exclusive or* $p \oplus q$ is True when exactly one of p and q is True and is False otherwise.

Example:

p : Today is Friday.

q : It is raining today.

$p \oplus q$: **Either** today is Friday **or** it is raining today.

It is False on Friday when it is raining.

It is True on Friday when it is not raining.

It is True on days other than Friday when it is raining.

It is False on days other than Friday when it is not raining.

Table 4: The Truth Table for the Exclusive OR of Two Propositions.

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Conditional Statements

Definition 5: Implication (IF-THEN):

Implication or conditional operator ($\rightarrow$): The implication or conditional statement $p \rightarrow q$ is the proposition “if p , then q .” It is False when p is true and q is false, and True otherwise.

p is called the *hypothesis* (or *antecedent* or *premise*).

q is called the *conclusion* (or *consequence*)

Example 7:

p : Matin learns CSE.

q : Matin will find a good job.

$p \rightarrow q$: **If** Matin learns CSE, **then** he will find a good job.

It is True if Matin learns CSE and finds a good job.

It is False if Matin learns CSE and does not find a good job.

It is True if Matin does not learn CSE and finds a good job.

It is True if Matin does not learn CSE and does not find a good job.

Table 5: The Truth Table for the Implication $p \rightarrow q$.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

CONVERSE, CONTRAPOSITIVE, AND INVERSE

Implication: $p \rightarrow q$

Converse of $p \rightarrow q$: $q \rightarrow p$

Contrapositive of $p \rightarrow q$: $\neg q \rightarrow \neg p$

Inverse of $p \rightarrow q$: $\neg p \rightarrow \neg q$

Equivalent Compound Propositions:

When two compound propositions always have the same truth value, then they are **equivalent**.

An implication and its contrapositive are equivalent, that is, $p \rightarrow q \equiv \neg q \rightarrow \neg p$.

p	q	$\neg q$	$\neg p$	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	F	F	T	T
T	F	T	F	F	F
F	T	F	T	T	T
F	F	T	T	T	T

The converse and the inverse of an implication are equivalent, that is, $q \rightarrow p \equiv \neg p \rightarrow \neg q$.

p	q	$\neg p$	$\neg q$	$q \rightarrow p$	$\neg p \rightarrow \neg q$
T	T	F	F	T	T
T	F	F	T	T	T
F	T	T	F	F	F
F	F	T	T	T	T

Example 9:

What are the contrapositive, the converse, and the inverse of the following conditional statement?

The home team wins whenever it is raining.

Solution:

The original statement can be rewritten as

If it is raining, then the home team wins. ($p \rightarrow q$)

The contrapositive is

If the home team does not win, then it is not raining. ($\neg q \rightarrow \neg p$)

The converse is

If the home team wins, then it is raining. ($q \rightarrow p$)

The inverse is

If it is not raining, then the home team does not win. ($\neg p \rightarrow \neg q$)

BI-CONDITIONALS

Definition 6:

Bi-implication or bi-conditional or equivalence operator ($\leftrightarrow$): The bi-implication or bi-conditional or equivalence statement $p \leftrightarrow q$ is the proposition “if and only if p , then q ”. It is True when p and q have the same truth values, and is False otherwise.

Table 6: The Truth Table for the Bi-implication $p \leftrightarrow q$.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

The bi-implication “if and only if p , then q ” is equivalent to “if p , then q and if q , then p ”.

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

p	q	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

Example 10:

p : You can take the flight.

q : You buy a ticket.

$p \leftrightarrow q$: You can take the flight if and only if you buy a ticket.

If you took the flight, then you bought a ticket and if you bought a ticket, then you took the flight.

Truth Tables of Compound propositions**Example 11:**

Construct the truth table of the compound proposition $(p \vee \neg q) \rightarrow (p \wedge q)$.

Table 7: The Truth Table of $(p \vee \neg q) \rightarrow (p \wedge q)$

$p \ q$	$\neg q$	$p \vee \neg q$	$p \wedge q$	$(p \vee \neg q) \rightarrow (p \wedge q)$
T T	F	T	T	T
T F	T	T	F	F
F T	F	F	F	T
F F	T	T	F	F

Precedence of Logical Operations

Table 8: Precedence of Logical Operators.

Operator	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

$\neg p \wedge q$ means $(\neg p) \wedge q$

$p \wedge q \vee r$ means $(p \wedge q) \vee r$

$p \vee q \rightarrow r$ means $(p \vee q) \rightarrow r$

Translating English Sentences

Example 12:

You can access the Internet from campus only if you are a computer science major or you are not a freshman.

Let

a : You can access the Internet from campus.

c : You are a computer science major.

f : You are freshman.

The sentence can be represented as

$$a \rightarrow (c \vee \neg f)$$

Example 13:

You cannot ride the roller coaster if you are under 4 feet tall unless you are older than 16 years old.

Let

q : You can ride the roller coaster.

r : You are under 4 feet tall.

s : You are older than 16 years old.

The sentence can be translated to

$$(r \wedge \neg s) \rightarrow \neg q$$

1.2 Propositional Equivalences

Definition 1:

A compound proposition that is always true is called a *tautology*.

A compound proposition that is always false is called a *contradiction*.

A proposition that is neither a tautology nor a contradiction is called a *contingency*.

Example 1:

Table 1: Examples of a Tautology and a Contradiction.

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

$p \vee \neg p$ is a tautology

$p \wedge \neg p$ is a contradiction

Logical Equivalences

Compound propositions that have the same truth values in all possible cases are called **logically equivalent**.

The notation $p \equiv q$ denotes that p and q are logically equivalent.

Table 2: De Morgan's Laws.

$$\neg (p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg (p \vee q) \equiv \neg p \wedge \neg q$$

De Morgan's laws extend to

$$\neg (p_1 \vee p_2 \vee \dots \vee p_n) \equiv (\neg p_1 \wedge \neg p_2 \wedge \dots \wedge \neg p_n)$$

$$\neg (p_1 \wedge p_2 \wedge \dots \wedge p_n) \equiv (\neg p_1 \vee \neg p_2 \vee \dots \vee \neg p_n)$$

Example 2: Show that $\neg(p \vee q) \equiv \neg p \wedge \neg q$.

Table 3: Truth Tables for showing $\neg(p \vee q) \equiv \neg p \wedge \neg q$.

$p \ q$	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T T	T	F	F	F	F
T F	T	F	F	T	F
F T	T	F	T	F	F
F F	F	T	T	T	T

From the truth table we see that $\neg(p \vee q) \equiv \neg p \wedge \neg q$.

Example 3: Show that $p \rightarrow q \equiv \neg p \vee q$.

Table 4: Truth Tables for showing $p \rightarrow q \equiv \neg p \vee q$.

$p \ q$	$\neg p$	$\neg p \vee q$	$p \rightarrow q$
T T	F	T	T
T F	F	F	F
F T	T	T	T
F F	T	T	T

From the truth table we see that $p \rightarrow q \equiv \neg p \vee q$.

Example 4: Show that $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ [Distributive law].

Table 5: A Demonstration that $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.

$p \ q \ r$	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T T T	T	T	T	T	T
T T F	F	T	T	T	T
T F T	F	T	T	T	T
T F F	F	T	T	T	T
F T T	T	T	T	T	T
F T F	F	F	T	F	F
F F T	F	F	F	T	F
F F F	F	F	F	F	F

From the truth table we see that $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.

1.3 Predicates and Quantifiers

Predicates

- The statement “ x is greater than 3” has two parts. The variable x is the subject of the statement. “is greater than 3” is the **predicate**, which refers to a property that the subject of the statement can have.
- We denote the statement “ x is greater than 3” by $P(x)$, where **P denotes the predicate “is greater than 3”** and **x is the variable**.
- The statement **$P(x)$** is said to be the value of the **propositional function P** at x .
- Once a value has been assigned to the variable x , the statement $P(x)$ becomes a proposition and has a truth value.

Example 1: Let $P(x): x > 3$. What are the truth values of $P(4)$ and $P(2)$?

Solution:

$P(4): 4 > 3$ is True.

$P(2): 2 > 3$ is False.

We can denote the statement “ $x = y + 3$ ” by $Q(x, y)$, where x and y are variables and Q is the predicate. When values are assigned to the variables x and y , the statement $Q(x, y)$ has a truth value.

Example 3: Let $Q(x, y): x = y + 3$. What are the truth values of $Q(1, 2)$ and $Q(3, 0)$?

Solution:

$Q(1, 2): 1 = 2 + 3$ is False.

$Q(3, 0): 3 = 0 + 3$ is True.

We can let $R(x, y, z)$ denote the statement “ $x + y = z$ ”. When values are assigned to the variables x , y , and z , this statement has a truth value.

Example 5: Let $R(x, y, z): x + y = z$. What are the truth values of $R(1, 2, 3)$ and $R(0, 0, 1)$?

Solution:

$R(1, 2, 3): 1 + 2 = 3$ is True.

$R(0, 0, 1): 0 + 0 = 1$ is False.

Quantifiers

- **Quantification** is a way of creating a proposition from a propositional function.
- The area of logic that deals with predicates and quantifiers is called the **predicate calculus**.

THE UNIVERSAL QUANTIFIER

- All possible values of a variable in a particular domain are called the **domain of discourse** or **universe of discourse** or the **domain**.
- The universal quantification of $P(x)$ for a particular domain is the proposition that asserts that $P(x)$ is true for all values of x in this domain.

Definition 1: The *universal quantification* of $P(x)$ is the statement “ $P(x)$ for all values of x in the domain.” The notation $\forall xP(x)$ denotes the universal quantification of $P(x)$. Here $\forall$ is called the *universal quantifier*.

We read $\forall xP(x)$ as “for all x $P(x)$ ” or “for every x $P(x)$ ”

An element for which $P(x)$ is false is called a **counterexample** of $\forall xP(x)$.

Example 8: Let $P(x): x + 1 > x$. What is the truth value of the quantification $\forall xP(x)$, where the domain consists of all real numbers?

Solution:

Because $P(x)$ is true for all real numbers x , the quantification $\forall xP(x)$ is true.

Example 9: Let $Q(x): x < 2$. What is the truth value of the quantification $\forall xQ(x)$, where the domain consists of real numbers?

Solution:

If $x = 3$, then $(x < 2) = (3 < 2)$ is not true (counterexample). Therefore, $\forall xQ(x)$ is false.

Example 10: Let $P(x): x^2 > 0$. What is the truth value of the quantification $\forall xP(x)$, where the domain consists of integers?

Solution:

If $x = 0$, then $(x^2 > 0) = (0^2 > 0)$ is not true (counterexample). Therefore, $\forall xP(x)$ is false.

When all the elements in the domain can be listed, say $x_1, x_2, \dots, x_n$, it follows that the universal quantification $\forall xP(x)$ is the same as the conjunction
$$\forall xP(x) \equiv P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n)$$

Example 11: What is the truth value of $\forall xP(x)$, where $P(x)$ is the statement “ $x^2 < 10$ ” and the domain consists of the positive integers not exceeding 4?

Solution:

Domain = $\{1, 2, 3, 4\}$

$$\forall xP(x) \equiv P(1) \wedge P(2) \wedge P(3) \wedge P(4) \equiv T \wedge T \wedge T \wedge F \equiv F$$

Therefore, $\forall xP(x)$ is false.

Example 13: What is the truth value of $\forall x(x^2 \geq x)$ if the domain consists of all real numbers? What is the truth value of this statement if the domain consists of all integers?

Solution:

$$x^2 \geq x$$

$$\text{or, } x^2 - x \geq 0$$

$$\text{or, } x(x - 1) \geq 0.$$

Consequently, $x^2 \geq x$ if and only if $x \leq 0$ or $x \geq 1$.

If the domain consists of all real numbers,

$\forall x(x^2 \geq x)$ is false (since $x^2 \geq x$ is false for all real numbers x with $0 < x < 1$).

If the domain consists of integers,

$\forall x(x^2 \geq x)$ is true (since there are no integers x with $0 < x < 1$).

THE EXISTENTIAL QUANTIFIER

With existential quantification, we form a proposition that is true if and only if $P(x)$ is true for at least one value of x in the domain.

Definition 2: The *existential quantification* of $P(x)$ is the proposition “There exists at least one element x in the domain such that $P(x)$.” We use the notation $\exists xP(x)$ for the existential quantification of $P(x)$. Here $\exists$ is called the *existential quantifier*.

Example 14: Let $P(x)$ denote the statement “ $x > 3$.” What is the truth value of the quantification $\exists xP(x)$, where the domain consists of all real numbers?

Solution:

Domain = All real numbers

Since $P(4)$: $4 > 3$ is true, $\exists xP(x)$ is true (example).

Example 15: Let $Q(x)$ denote the statement “ $x = x + 1$.” What is the truth value of the quantification $\exists xQ(x)$, where the domain consists of all real numbers?

Solution:

Domain: All real numbers

Since $Q(x)$ is false for every real number x , $\exists xQ(x)$ is false.

When all the elements in the domain can be listed, say $x_1, x_2, \dots, x_n$, the existential quantification $\exists xP(x)$ is the same as the disjunction

$$\exists xP(x) \equiv P(x_1) \vee P(x_2) \vee \dots \vee P(x_n).$$

Example 16: What is the truth value of $\exists xP(x)$, where $P(x)$ is the statement “ $x^2 > 10$ ” and the domain consists of the positive integers not exceeding 4?

Solution:

Domain = $\{1, 2, 3, 4\}$

$$\exists xP(x) \equiv P(1) \vee P(2) \vee P(3) \vee P(4) \equiv F \vee F \vee F \vee T \equiv T$$

Therefore, $\exists xP(x)$ is true.

Precedence of Quantifiers

The quantifiers $\forall$ and $\exists$ have higher precedence than all logical operators from propositional calculus.

For example, $\forall xP(x) \vee Q(x)$ means $(\forall xP(x)) \vee Q(x)$.

Binding Variables

- When a quantifier is used on the variable x , we say that this occurrence of the variable is **bound**.
- An occurrence of a variable that is not bound by a quantifier or set equal to a particular value is said to be **free**.
- All the variables that occur in a propositional function must be bound or set equal to a particular value to turn it into a proposition.
- The part of a logical expression to which a quantifier is applied is called the **scope** of this quantifier.
- A variable is free if it is outside the scope of all quantifiers in the formula that specifies this variable.

Example 18:

In the statement $\exists x(x + y = 1)$, the variable x is bound by the existential quantification $\exists x$, but the variable y is free because it is not bound by a quantifier and no value is assigned to this variable.

Example:

In the statement $\exists x(P(x) \wedge Q(x)) \vee \forall xR(x)$, all occurrences of the variable x are bound.

The scope of the first quantifier, $\exists x$, is the expression $P(x) \wedge Q(x)$.

The scope of the second quantifier, $\forall x$, is the expression $R(x)$.

Negating Quantified Expressions

Consider the proposition

“Every student in the class has taken a course in calculus.”

This statement is a universal quantification $\forall x P(x)$

where $P(x)$ is the statement “ x has taken a course in calculus.”

The negation of this statement is

“There is at least one student in the class who has not taken a course in calculus.”

This is simply the existential quantification of the negation of the original propositional function $\exists x \neg P(x)$.

From this illustration we have the following equivalence:

$$\neg \forall x P(x) \equiv \exists x \neg P(x).$$

Consider the proposition

“There is at least one student in this class who has taken a course in calculus.”

This is the existential quantification $\exists xQ(x)$

where $Q(x)$ is the statement “ x has taken a course in calculus.”

The negation of this statement is the proposition

“Every student in this class has not taken calculus,”

This is just the universal quantification of the negation of the original propositional function $\forall x \neg Q(x)$.

This example illustrates the equivalence

$$\neg \exists x Q(x) \equiv \forall x \neg Q(x).$$

Example 21: What are the negations of the statements $\forall x(x^2 > x)$ and $\exists x(x^2 = 2)$?

Solution:

$$\begin{aligned} &\neg \forall x(x^2 > x) \\ &\equiv \exists x \neg (x^2 > x) \\ &\equiv \exists x (x^2 \leq x). \end{aligned}$$

$$\begin{aligned} &\neg \exists x(x^2 = 2) \\ &\equiv \forall x \neg (x^2 = 2) \\ &\equiv \forall x(x^2 \neq 2). \end{aligned}$$

Example 22: Show that $\neg \forall x(P(x) \rightarrow Q(x))$ and $\exists x(P(x) \wedge \neg Q(x))$ are logically equivalent.

Solution:

$$\begin{aligned} &\neg \forall x(P(x) \rightarrow Q(x)) \\ &\equiv \exists x(\neg(P(x) \rightarrow Q(x))) \\ &\equiv \exists x(\neg(\neg P(x) \vee Q(x))) \\ &\equiv \exists x(\neg(\neg P(x)) \wedge \neg Q(x)) \\ &\equiv \exists x(P(x) \wedge \neg Q(x)) \end{aligned}$$

1.4 Nested Quantifiers

Nested quantifiers are quantifiers that occur within the scope of other quantifiers, such as in the statement $\forall x \exists y (x + y = 0)$.

Example 1:

Assume that the domain for the variables x and y consists of all real numbers.

The statement $\forall x \forall y (x + y = y + x)$ says that $x + y = y + x$ for all real numbers x and y (commutative law for addition of real numbers).

The statement $\forall x \exists y (x + y = 0)$ says that for every real numbers x there is a real number y such that $x + y = 0$, which states that every real number has an additive inverse.

The statement $\forall x \forall y \forall z (x + (y + z) = (x + y) + z)$ is the associative law for addition of real numbers.

The Order of Quantifiers

Example 3: Let $P(x, y)$ be the statement “ $x + y = y + x$.” What are the truth values of the quantifications $\forall x \forall y P(x, y)$ and $\forall y \forall x P(x, y)$ where the domain for all variables consists of all real numbers?

Solution:

The quantification $\forall x \forall y P(x, y)$ denotes the proposition
“For all real numbers x , for all real numbers y , $x + y = y + x$.”

Because $P(x, y)$ is true for all real numbers x and y , the proposition $\forall x \forall y P(x, y)$ is true.

The statement $\forall y \forall x P(x, y)$ says
“For all real numbers y , for all real numbers x , $x + y = y + x$.”

Because $P(x, y)$ is true for all real numbers y and x , the proposition $\forall y \forall x P(x, y)$ is true.

Example 4: Let $Q(x, y)$ denote “ $x + y = 0$.” What are the truth values of the quantifications $\exists y \forall x Q(x, y)$ and $\forall x \exists y Q(x, y)$, where the domain for all variables consists of real numbers?

Solution:

The quantification $\exists y \forall x Q(x, y)$ denotes the proposition
“There is a real number y such that for every real number x , $Q(x, y)$.”

Since there is no real number y such that $x + y = 0$ for all real numbers x , the statement $\exists y \forall x Q(x, y)$ is false.

The quantification $\forall x \exists y Q(x, y)$ denotes the proposition
“For every real number x there is a real number y such that $Q(x, y)$ ”.

Given a real number x , there is a real number y (namely $y = -x$) such that $x + y = 0$. Hence, the statement $\forall x \exists y Q(x, y)$ is true.

Example 5: Let $Q(x, y, z)$ be the statement “ $x + y = z$.” What are the truth values of the statement $\forall x \forall y \exists z Q(x, y, z)$ and $\exists z \forall x \forall y Q(x, y, z)$, where the domain of all variables consist of all real numbers?

Solution:

Suppose that x and y are assigned values. Then, there exists a real number such that $x + y = z$. Thus, $\forall x \forall y \exists z Q(x, y, z)$ is true.

There is no value of z that satisfies the equation $x + y = z$ for all values of x and y . Thus $\exists z \forall x \forall y Q(x, y, z)$ is false.

Negating Nested Quantifiers

Example 14: Express the negation of the statement $\forall x \exists y (xy = 1)$ so that no negation precedes a quantifier.

Solution:

$$\begin{aligned} & \neg \forall x \exists y (xy = 1) \\ & \equiv \exists x \neg \exists y (xy = 1) \\ & \equiv \exists x \forall y \neg (xy = 1) \\ & \equiv \exists x \forall y (xy \neq 1) \end{aligned}$$

1.6 Introduction to proofs

A proof is a valid argument that establishes the truth of a mathematical statement.

Direct Proofs

A **direct proof** shows that a conditional statement $p \rightarrow q$ is true by showing that if p is true, then q must also be true.

Definition 1: The integer n is *even* if there exists an integer k such that $n = 2k$, and it is *odd* if there exists an integer k such that $n = 2k + 1$.

Example 1: Give a direct proof of the theorem “If n is an odd integer, then n^2 is an odd integer.”

Solution:

Assume that the hypothesis of this conditional statement is true, namely, assume that n is odd.

Then $n = 2k + 1$, where k is an integer.

It follows that

$$\begin{aligned} n^2 &= (2k + 1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1 \\ &= 2m + 1, \text{ where } m = 2k^2 + 2k \text{ is an integer} \end{aligned}$$

So, n^2 is an odd integer.

Therefore, the given theorem is proved.

Example 2: Give a direct proof that “If m and n are both perfect squares, then mn is also a perfect square.” (An integer a is a **perfect square** if there is an integer b such that $a = b^2$.)

Solution:

Assume that m and n are perfect squares.

Then $m = s^2$ and $n = t^2$, where s and t are integers.

It follows that

$mn = s^2 t^2 = (st)^2 = k^2$, where $k = st$ is an integer.

So, mn is perfect square.

Therefore, the given theorem is proved.

Proofs by Contraposition

Proofs by Contraposition make use of the fact that the conditional statement $p \rightarrow q$ is equivalent to its contrapositive, $\neg q \rightarrow \neg p$.

This means that the conditional statement $p \rightarrow q$ can be proved by showing that its contrapositive, $\neg q \rightarrow \neg p$, is true.

Example 3: Prove that “If $3n + 2$ is odd, then n is odd,” where n is an integer.

Solution:

Implication: If $3n + 2$ is odd, then n is odd.

Contrapositive: If n is even, then $3n + 2$ is even.

Assume that n is even.

Then $n = 2k$ for some integer k .

It follows that

$3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1) = 2m$, where $m = 3k + 1$ is an integer.

So, $3n + 2$ is even.

As the contrapositive is true, the original implication is true.

Example 4: Prove that “If $n = ab$, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$,” where a and b are positive integers.

Solution:

Implication: If $n = ab$, then $(a \leq \sqrt{n}) \vee (b \leq \sqrt{n})$

Contrapositive: If $\neg ((a \leq \sqrt{n}) \vee (b \leq \sqrt{n}))$, then $n \neq ab$

Assume that $\neg ((a \leq \sqrt{n}) \vee (b \leq \sqrt{n}))$ is true.

Then

$$\begin{aligned} & \neg ((a \leq \sqrt{n}) \vee (b \leq \sqrt{n})) \\ & \equiv \neg (a \leq \sqrt{n}) \wedge \neg (b \leq \sqrt{n}) \\ & \equiv (a > \sqrt{n}) \wedge (b > \sqrt{n}) \end{aligned}$$

This shows that

$$ab > \sqrt{n} \cdot \sqrt{n} = n$$

Therefore, $n \neq ab$

As the contrapositive is true, the original implication is true.

Proofs by Contradiction

The statement $(r \wedge \neg r)$ is a contradiction, where r is any proposition.

We can prove that a proposition p is true if we can show that $\neg p \rightarrow (r \wedge \neg r)$ is true.

Example 10: Prove that “ $\sqrt{2}$ is irrational” by giving a proof by contradiction.

Solution:

Suppose that $\sqrt{2}$ is rational. Then, $\sqrt{2} = a/b$, where a and b are integers and have no common factors.

$$\begin{aligned}\sqrt{2} &= a/b \\ \text{or, } 2 &= a^2/b^2 \\ \text{so, } 2b^2 &= a^2\end{aligned}$$

It follows that a^2 is even and a must also be even. Then, $a = 2c$ for some integer c .

$$\begin{aligned}\text{Thus,} \\ 2b^2 &= a^2 = 4c^2 \\ \text{so, } b^2 &= 2c^2\end{aligned}$$

This means that b^2 is even and b must be even. Then, $b = 2d$ for some integer d .

$$\text{Now, } \sqrt{2} = a/b = (2c)/(2d)$$

Thus, a and b have a common factor 2, which contradicts with the assumption that a and b have no common factors.

Therefore, $\sqrt{2}$ is not rational, that means, $\sqrt{2}$ is irrational.

Example 11: Give a proof by contradiction of the theorem “If $3n + 2$ is odd, then n is odd.”

Solution:

Assume that “If $3n + 2$ is odd, then n is even.”

Then $n = 2k$ for some integer k .

It follows that

$$\begin{aligned} 3n + 2 &= 3(2k) + 2 \\ &= 6k + 2 \\ &= 2(3k + 1) \end{aligned}$$

So, $3n + 2$ is even. This contradicts the assumption that $3n + 2$ is odd.

Therefore, “If $3n + 2$ is odd, then n is even” is not true.

This implies that “If $3n + 2$ is odd, then n is odd” is true.

Chapter 4

Induction and Recursion

4.1 Mathematical Induction

Mathematical induction is used to prove statements that assert that $P(n)$ is true for all nonnegative integers n , where $P(n)$ is a propositional function.

PRINCIPLE OF MATHEMATICAL INDUCTION

To prove that $P(n)$ is true for all nonnegative integers n , where $P(n)$ is a propositional function, we complete two steps:

BASIS STEP: We verify that $P(0)$ or $P(1)$ (which one is the first element of the domain) is true.

INDUCTIVE STEP: We show that the conditional statement $P(k) \rightarrow P(k + 1)$ is true for all nonnegative integers k .

To complete the inductive step of the proof, we assume that $P(k)$ is true for an arbitrary nonnegative integer k and show that under this assumption, $P(k + 1)$ must also be true. The assumption that $P(k)$ is true is called the **inductive hypothesis**.

Once we complete both steps in a proof by mathematical induction, we have shown that $P(n)$ is true for all nonnegative integers, that is, we have shown that $\forall n P(n)$ is true where the quantification is over the set of nonnegative integers.

Example 1: Show that if n is a positive integer, then $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$.

Solution:

Let $P(n)$: $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$.

BASIS STEP: Let $n = 1$

LHS = 1 and RHS = $\frac{1(1+1)}{2} = 1$. As LHS = RHS, $P(1)$ is true.

INDUCTIVE STEP: Assume that $P(k)$ holds for an arbitrary positive integer k . That is, we assume that $P(k)$: $1 + 2 + \cdots + k = \frac{k(k+1)}{2}$. (1)

Under this assumption, it must be shown that $P(k + 1)$ is true.

Adding $(k + 1)$ at both the sides of (1)

$$1 + 2 + \cdots + k + (k + 1) = \frac{k(k+1)}{2} + (k + 1) = \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+1)(k+2)}{2} = \frac{(k+1)[(k+1)+1]}{2} = P(k + 1)$$

Therefore, $P(k) \rightarrow P(k + 1)$ is true.

Since $P(1)$ is true and $P(k) \rightarrow P(k + 1)$ is true for all positive integers k , $P(n)$ is true for all positive integers n .

Example 2: Prove that the sum of the first n odd positive integers is n^2 .

Solution:

Let $P(n)$: $1 + 3 + 5 + \dots + (2n - 1) = n^2$.

BASIS STEP: Let $n = 1$

LHS = 1 and RHS = $1^2 = 1$. As LHS = RHS, $P(1)$ is true.

INDUCTIVE STEP: Assume that $P(k)$ holds for an arbitrary odd positive integer k . That is, we assume that $P(k)$: $1 + 3 + 5 + \dots + (2k - 1) = k^2$. (1)

Under this assumption, it must be shown that $P(k + 1)$ is true.

Adding $[2(k + 1) - 1]$ at both the sides of (1)

$$1 + 3 + 5 + \dots + (2k - 1) + [2(k + 1) - 1] = k^2 + [2(k + 1) - 1] = k^2 + 2k + 2 - 1 = k^2 + 2k + 1 = (k + 1)^2 = P(k + 1)$$

Therefore, $P(k) \rightarrow P(k + 1)$ is true.

Since $P(1)$ is true and $P(k) \rightarrow P(k + 1)$ is true for all odd positive integers k , $P(n)$ is true for all odd positive integers n .

Example 3: Use mathematical induction to show that $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$ for all nonnegative integers n .

Solution:

Let $P(n)$: $1 + 2 + 2^2 + \dots + 2^n = 2^0 + 2^1 + 2^2 + \dots + 2^n = 2^{n+1} - 1$.

BASIS STEP: Let $n = 0$

LHS = $2^0 = 1$ and RHS = $2^{0+1} - 1 = 1$. As LHS = RHS, $P(0)$ is true.

INDUCTIVE STEP: Assume that $P(k)$ is true, that is, we assume that

$$P(k): 2^0 + 2^1 + 2^2 + \dots + 2^k = 2^{k+1} - 1 \quad (1)$$

Under this assumption, it must be shown that $P(k + 1)$ is true.

Adding 2^{k+1} at both the sides of (1)

$$2^0 + 2^1 + 2^2 + \dots + 2^k + 2^{k+1} = 2^{k+1} - 1 + 2^{k+1} = 2 \cdot 2^{k+1} - 1 = 2^{(k+1)+1} - 1 = P(k + 1)$$

Therefore, $P(k) \rightarrow P(k + 1)$ is true.

Since $P(0)$ is true and $P(k) \rightarrow P(k + 1)$ is true for all nonnegative integers k , $P(n)$ is true for all nonnegative integers n .